

Technical Executive Summary

Pricing American Options Using the Least Squares Monte Carlo (LSM) Method

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1. Project Overview & Objective

This project traces a complete engineering arc: from the theoretical mathematics of American option pricing to a deployed, multi-asset institutional risk management platform. The central challenge is that American options—unlike their European counterparts—grant the holder the right to exercise at *any* point before maturity T , transforming pricing into an **optimal stopping problem** that no closed-form formula such as Black-Scholes-Merton can solve.

The solution is the **Longstaff-Schwartz LSM algorithm**, which approximates the continuation value at every discrete time node via cross-sectional regression over simulated paths, resolving the exercise boundary through backward induction. This core was then stress-tested against real 2025 market data, upgraded with two advanced stochastic architectures (Merton Jump-Diffusion and Markov Regime-Switching), packaged into a commercial underwriting strategy, diversified across a multi-asset book, and deployed as a live Python web application.

2. The Core Mathematical Engine

2.1 Asset Path Simulation: Geometric Brownian Motion

Under the risk-neutral measure, asset price dynamics follow $dS_t = rS_t dt + \sigma S_t dW_t$. Applying Itô's Lemma yields the exact time-step discretization:

$$S_{t_{k+1}} = S_{t_k} \exp \left[\left(r - \frac{\sigma^2}{2} \right) \Delta t + \sigma \sqrt{\Delta t} Z_k \right], \quad Z_k \sim \mathcal{N}(0, 1). \quad (1)$$

Parameters were derived empirically from five years of AAPL data (2020–2024): $S_0 = \$250.60$, $\hat{\sigma} = 31.66\%$ (Bessel-corrected log-returns scaled by $\sqrt{252}$), $r = 5\%$. A **10,000-path simulation matrix** of dimension $10,000 \times 51$ was generated via fully vectorized NumPy. The empirical ITM probability (50.36%) confirmed convergence to the theoretical limit (50.02%), validating statistical integrity.

2.2 The Longstaff-Schwartz Backward Induction

For an ATM American put with $K = S_0 = \$250.60$, the intrinsic payoff is $H_{i,j} = \max(K - S_{i,j}, 0)$, initialized at maturity as $V_{i,M} = H_{i,M}$. The per-step discount factor is $\gamma = e^{-r\Delta t}$. At each backward step j :

1. **Filter** to strictly ITM paths ($H_{i,j} > 0$); OTM paths trivially continue.
2. **Regress** discounted future values $Y_i = V_{i,j+1} \cdot \gamma$ against current prices via a **2nd-degree polynomial basis**:

$$\hat{C}_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2. \quad (2)$$

3. **Exercise rule:** If $H_{i,j} > \hat{C}_i$, record immediate exercise and zero out all downstream cash flows. Otherwise:

$$V_{i,j} = V_{i,j+1} \cdot \gamma.$$

The American put price is the mean discounted value across all paths. The **Early Exercise Premium** = $P_{\text{American}} - P_{\text{European}}$ is isolated using the identical simulated matrix, eliminating Monte Carlo variance entirely.

3. Stress-Testing & Behavioral Risk

3.1 The 2025 Historical Backtest

The LSM engine and its stored regression boundaries β_t were validated against the actual out-of-sample 2025 AAPL trajectory. During a severe spring drawdown, the asset fell to \$198.85—generating \$51.75 of intrinsic value—yet the LSM continuation value dominated, blocking early exercise. AAPL subsequently recovered well above $K = \$250.60$, expiring OTM. The desk captured **100% of the initial premium**, confirming mathematical durability under live turbulence.

3.2 Stochastic Human Behavioral Attrition

The baseline assumes perfectly rational counterparties—a structural flaw. A **Stochastic Human Behavioral Attrition** model replaces synchronized mass exercise with 10,000 independent agents, each with a dynamic per-step exercise probability:

Intrinsic Value Zone	Behavioral Regime	Exercise Probability
$H_t < \$20$	Low inactivity — holders passive	0%
$\$20 \leq H_t < \45	Escalating temptation — staggered capitulation	$0.001 \times (H_t - 20)$
$H_t \geq \$45$	Panic threshold — stop-loss cascades	15% per step

A hard **Profit-Locking Threshold** $T_{\text{greed}} = \$45$ triggers exercise if either $H_t > \hat{C}_t$ (rational) or $H_t \geq T_{\text{greed}}$ (behavioral). During the 2025 crash at Step 13 (stock: \$203.70, payout: \$46.89), the behavioral trigger fired despite the regression predicting superior continuation, generating a corporate loss of \$205,803. This proves that **rational-agent assumptions create catastrophic unhedged liability** under real market conditions.

4. Advanced Stochastic Architectures

4.1 Merton Jump-Diffusion: Discontinuous Shock Events

Standard GBM cannot reproduce sudden price gaps inherent in financial crises. The **Merton Jump-Diffusion** model augments GBM with a compound Poisson process governed by:

- **Jump frequency** $\lambda = 1.5$: expected macroeconomic shocks per year.
- **Jump severity** ($\mu_J = -0.15$, $\sigma_J = 0.10$): a heavily negatively skewed log-normal shock distribution modeling asymmetric downside risk.

Running the full LSM algorithm over a fresh 10,000-path Merton matrix caused the American put premium to escalate from **\$26.31 (GBM) to \$32.49 (Merton)**—the \$6.18 differential precisely quantifying the excess capital buffer required against non-continuous market failures.

4.2 Markov Regime-Switching: Dynamic Macro State Calibration

Fixing $\sigma \approx 31.66\%$ uniformly ignores the empirical reality of alternating market cycles. The **Markov Regime-Switching** engine introduces a two-state classification:

- **Regime 0 (Expansionary):** compressed volatility σ_{low} , steady drift.
- **Regime 1 (Turbulent):** elevated volatility σ_{high} , erratic price action.

State transitions at each node are governed by the Markov Transition Matrix $P = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix}$, natively embedding macroeconomic cycles into the SDE. Because the asset spends the majority of its horizon in the stable expansionary regime, the Markov engine deflates the fair put value to **\$22.52**—demonstrating that blunt historical averaging structurally overprices options and destroys capital efficiency.

5. Commercial Strategy & Portfolio Diversification

5.1 The Loss-Leader Growth Strategy

With three model-derived fair values, a commercial underwriting strategy was designed to balance acquisition against balance-sheet protection:

- **Tranche 1 (Promotional / Markov):** First 1,000 units at \$22.52, functioning as aggressive loss-leader market capture during calm regimes.
- **Tranche 2 (Protective / Merton):** Remaining 9,000 units at \$32.49, providing the shock-absorbing buffer that keeps the desk solvent under tail events.

Stress-tested against the full granular attrition model, the protective tranche fully absorbed panic-driven outflows from the promotional tranche, confirming that **dynamic price blending strictly dominates uniform underwriting** at portfolio scale.

5.2 Modern Portfolio Theory: Multi-Asset Diversification

Asset	Role	Structural Function
AAPL	Core Tech Equity	Moderate-volatility premium baseline
NVDA	High-Beta Growth	Large upfront premium revenue engine
GLD	Safe-Haven Commodity	Inverse correlation hedge; absorbs equity panic liquidity
SPY	Diversified Macro Index	Low-beta anchor; suppresses aggregate book beta

Four independent LSM engines were synchronized to the 2025 crash timeline. When panic cascaded through

high-beta positions (AAPL, NVDA), the GLD and SPY hedges successfully absorbed the liquidity drain. The diversified book strictly outperformed single-asset underwriting—proving that intelligently weighted multi-asset options books can harvest high-volatility premiums while neutralizing catastrophic left-tail risk.

6. FinTech Deployment & Tech Stack

6.1 engine.py: The Live Pricing Backend

The backend module abstracts the full stochastic calculus pipeline into a low-latency pricing API with four core components:

1. **Live Data Ingestion:** Real-time spot price S_0 and a two-year historical window via the `yfinance` API.
2. **EWMA Volatility Calibration:** Rather than a static equally-weighted standard deviation, volatility is computed as:

$$\sigma_{\text{ewma}} = \sqrt{\text{EWMA}(R_t^2, \text{span} = 252) \times 252}, \quad R_t = \ln(P_t/P_{t-1}), \quad (3)$$

ensuring recent regimes carry dominant mathematical influence.

3. **BSM Pricing Core:** Standard Black-Scholes-Merton d_1 , d_2 , and $\Phi(\cdot)$ computed with $r = 4.3\%$ to produce baseline European premium P_{BSM} .
4. **Tiered Output Matrix:** Three commercial tiers operationalize the Loss-Leader strategy:

Tier	Label	Formula
A	Promo / Markov Proxy	$P_{\text{promo}} = 0.85 \times P_{\text{BSM}}$
B	Standard GBM	$P_{\text{std}} = 1.00 \times P_{\text{BSM}}$
C	Merton Shock Shield	$P_{\text{merton}} = 1.20 \times P_{\text{BSM}}$

Additional modules include **Dynamic Inventory Elasticity** (premiums scale with open interest to limit concentration risk) and an **Optimized Capacity Switching Threshold** (auto-flags the unit count at which acquisition pricing must pivot to Merton sizing).

6.2 app.py: The Institutional Streamlit Dashboard

The frontend is built in **Streamlit** with four key architectural features:

- **Master Routing & State Management:** `st.session_state` navigates across *Equity & Forecasting Views* (StockHome, Custom, Optimal) and *Options Underwriting Views* (CompanyOptions, CustomOptionsPortfolio).
- **Integration Plugs:** Decoupled UI-to-engine bridges trigger `LiveDataPricingEngine` on user input, rendering all three tiers with sub-second latency.
- **Aggregate Risk Ledger:** User selections compile into a centralized `pandas DataFrame` with real-time ROI tracking:

$$\text{ROI}_{\text{portfolio}} = \frac{\sum_{i=1}^N (\text{Predicted Sell}_i - \text{Buy Price}_i)}{\sum_{i=1}^N \text{Buy Price}_i} \times 100. \quad (4)$$

- **End-to-End Stress Tester:** Ingests any live ticker, constructs the full tiered product menu, and projects corporate survival margins against historical crash timelines.

Summary of Key Pricing Results

Architecture	Put Premium	Key Insight
Standard GBM + LSM	\$26.31	Theoretical fair value baseline
Merton Jump-Diffusion	\$32.49	+\$6.18 buffer against discrete fat-tail gaps
Markov Regime-Switching	\$22.52	−\$3.79 efficiency gain in calm-dominant markets
Loss-Leader Blend	\$22.52 / \$32.49	Optimal dual-tranche commercial structure

No single pricing abstraction is sufficient for institutional deployment. The mathematically optimal solution is a **Hybrid Framework** fusing Markov Regime-Switching for dynamic macro calibration with Merton Jump-Diffusion thresholds for fat-tail protection—delivered through a live, multi-asset FinTech platform that transforms academic stochastic calculus into real-time risk management infrastructure.